Essential in Max for engineers (40 Hours)

S.No	Session	Topics Covered	Type	Duration (hrs)
1	Session 1	**Introduction to Max** **Design Harmony** Max for Architects Autodesk 3ds Max **Terminologies** Modeling Lights and Cameras Animation, Rendering **The Welcome Screen** Learn Tab, Start Tab, Extend Tab Layout Panel **3ds Max Interface** Elements of GUI Application button and Quick Access toolbar Info Centre , Menu bar, Main toolbar Ribbon, Scene Explorer Viewport Layout Command panel Viewports Animation and Time control Viewport Navigation controls **Quad Menu** Transform Quadrant Display Quadrant **Tools Menu** Display Floater Array, Align Snapshot Color Clipboard Utility	Theory/Demo	2.0 Hrs
2	Session 2	Practice 01: Set viewports and assign styles as shown.	Practical	2.0 Hrs
3	Session 3	**Shapes and Geometry** **Spline**: Line, Rectangle, Circle, Ellipse Spline, Arc Spline, Donut Spline, NGon Spline, Star Spline, Helix Spline, Egg Spline, Section Spline, Extended Splines, WRectangle Spline, Channel Spline, Angle Spline, Tee Spline **Standard Primitives**: Box, Cone, Sphere, GeoSphere, Cylinder, Tube, Torus, Pyramid, Teapot, Plane, TextPlus **Extended Primitives**: Hedra, Torus Knot, Chamfer Box, ChamferCyl, Oil Tank, Capsule, Spindle, L-Ext, Gengon, C-Ext, Ring Wave, Prism, Hose **Compound Objects:** Morph, Scatter, Conform, Connect, BlobMesh, Shape Merge, Boolean, Terrain, Loft, Mesher, ProBoolean, ProCutter **Patch Grids**: Quad Patch, Tri Patch **NURBS Surfaces:** Point Surface , CV Surface **Doors:** Pivot Door, Sliding Door, BiFold Door **Windows:** Awning Window, Casement Window, Fixed Window, Pivoted Window, Projected Window, Sliding Window **AEC Extended:** Foliage, Railing, Wall **Stairs:** Straight Stair, L - Type Stair, U-Type Stair, Spiral Stair **Dynamics Object:** Spring, Damper **Space Warp Objects:** Gravity, Wind ,Displace	Theory/Demo	2.0 Hrs
4	Session 4	Practice 02: Model a sofa.	Practical	2.0 Hrs
5	Session 5	**Modifiers** **Selection Modifiers:** Mesh Select Modifier, Patch Select modifier, Poly Select Modifier Editable Poly **Vertex:** Remove, Break, Extrude, Weld, Chamfer, Target Weld **Edge:** Remove, Split, Extrude, Weld, Chamfer, Target Weld, Bridge, Connect **Border:** Extrude, Insert Vertex, Chamfer, Cap, Bridge, Connect **Polygon/Element:** Insert Vertex, Extrude, Outline, Bevel, Inset, Bridge, Flip	Theory/Demo	2.0 Hrs
6	Session 6	Practice 03: Model bedroom furniture set.	Practical	2.0 Hrs
		Modifiers		

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7	Session 7	**Object – Space Modifiers:** Bend Modifier, Lattice Modifier, Push Modifier, Shell Modifier, Stretch Modifier, Twist Modifier, Lathe Modifier, Wave Modifier, Smooth Modifier, TurboSmooth Modifier, Symmetry Modifier, Cloth Modifier, Simulation Parameters (Cloth)	Theory/Demo	2.0 Hrs
		FFD (Free Form Deformation)		
8	Session 8	Practice 04: Detail the bedroom furniture set using cloth and other modifiers.	Practical	2.0 Hrs
9	Session 9	**Modelling Interior Objects**	Theory/Demo	2.0 Hrs
		Detailed Modelling of a Kitchen Cabinet		
		Modelling the Objects for Living Room		
10	Session 10	Practice 05: Model the kitchen cabinet.	Practical	2.0 Hrs
11	Session 11	**Modelling Exterior Objects**	Theory/Demo	2.0 Hrs
		Modeling the Exterior of a Building		
		Environment Modeling		
12	Session 12	Practice 06: Model the exterior of a building.	Practical	2.0 Hrs
13	Session 13	**Materials**	Theory/Demo	2.0 Hrs
		Material Editor		
		Compact Material Editor: Get Material, Put Material to Scene ,Assign Material to Selection , Reset Map/Mtl to Default Settings, Make Material Copy, Make Unique, Put to Library, Material ID Channel flyout, Show Shaded/Realistic Material in Viewport flyout		
		Show End Result, Go to Parent, Go Forward to Sibling		
		Slate Material Editor: Menu bar, Toolbar, Material/Map Browser, Status, Active View, View navigation, Parameter Editor		
		UVW Coordinates: Coordinates Rollout (2D), Coordinates Rollout (3D), Camera Map Modifier (World Space)		
		UVW Map Modifier: Manipulators for UVW Map		
		Composite Material		
		Defining Material for Exterior Model		
14	Session 14	Practice 07: Create and assign materials for exterior model.	Practical	2.0 Hrs
15	Session 15	**Lights, Camera and Rendering**	Theory/Demo	2.0 Hrs
		Photometric Lights: Target Light, Free Light, Mental Ray (mr) Skylight Portal, Sun Positioner		
		Standard Lights: Target Spotlight, Free Spotlight, Target Directional Light, Free Directional Light, Omni Light, Skylight, mr Area Omni Light, mr Area Spotlight		
		Cameras: Physical Camera, Free Camera, Target Camera, Walkthrough Assistant		
16	Session 16	Practice 08: Make the model render ready- for interior render output.	Practical	2.0 Hrs
17	Session 17	**Lights, Camera and Rendering**	Theory/Demo	2.0 Hrs
		Rendering: Render Setup Dialog, ART Renderer, Scanline Renderer, Mental ray Renderer, Environment and Effects Dialog Box		
		Exposure Controls		
		Exterior Lighting		
		Environment Texturing		
		Exterior Rendering		
18	Session 18	Practice 09: Make the model render ready- exterior render output.	Practical	2.0 Hrs
19	Session 19	**Animation**	Theory/Demo	2.0 Hrs
		Saving and Loading Animation, Animation Layers (Layer Controller), Enabling Animation Layers, Animation and Time Controls, Integrating Animation Layers in Workflow, Time Configuration		
		Animation Quad Menu: Set quadrant, Pose quadrant, Transform quadrant		
		Animation Constraints, Wire Parameters, Position/Rotation/Scale Controller, Video Post		
		Day Light Shadow Animation		
		Chess Animation		
		Files Management		
		File – Handling: New, Save, Merge, Manage, Properties, Exit		
		Merging Files		
		About the Print Size Wizard		
		Project Worksheet		
20	Session 20	Practice 10: Create an animation and export it.	Practical	2.0 Hrs